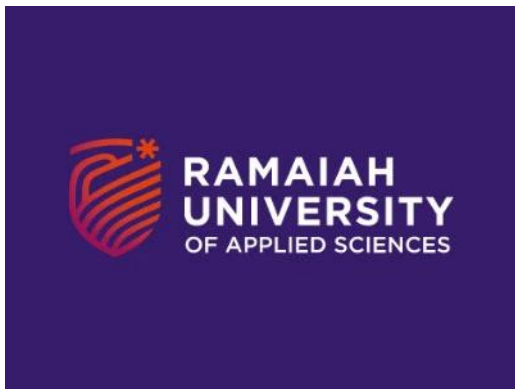
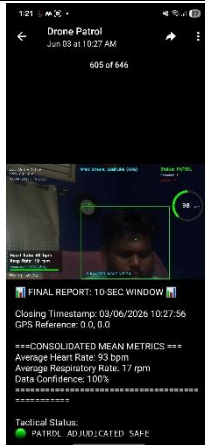
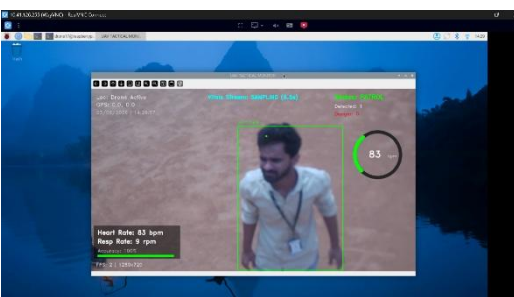


Autonomous UAV System For Real-Time Human Detection and Health Vitals Monitoring		
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Keywords: Vitals Monitoring System, Drone-Based Search and Rescue, Victim Detection, Emergency Response Systems, Real-Time Health Monitoring.		
Abstract: Through this project, a Vitals Monitoring System for the drone equipped with an Autonomous Contactless Monitoring Platform to monitor the person in the disaster-affected zone to deliver real-time information to the Search and rescue teams . the system is a working prototype that will help rescue teams in finding victims; in assessing the victim's condition at a distance; and in reducing danger when approaching the individual for face-to-face assistance in dangerous situations. The system would therefore provide an efficient and effective response in the case of emergencies such as autonomous navigation, victim detection and real time health monitoring. The results of the project indicates that the proposed system can detect human body, approach to the victim, gather health information without touching the victim. The system maintained an impressive 91.35% accuracy at a 1-meter range, which further scaled up to 91.76% accuracy at 2 meters as the frame stabilized. Environmental Adaptability: Under optimal, direct light conditions, the model achieved a peak accuracy of 93.61% with a minimal percentage error of 6.39%. Stress Resilience: Even under challenging adverse conditions such as harsh backlighting (84.27% accuracy) and physical target sweat/moisture conditions (84.60% accuracy)		
   		
Figure (a) Mobile Interface Figure (b): Telegram Output Figure (c): Hardware Prototype (Hexacopter Drone) Figure (d): heart rate detection from drone (2.5 meters)		

